



EXECUTIVE INSIGHT

LOGISTICS OPERATIONS PERFORMANCE REVIEW (2022 – 2024)

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Executive Summary

The logistics operation maintained **stable operational performance** throughout the three-year period. Fleet utilization remained around 83%, revenue consistently exceeded operating costs, and fuel efficiency showed minimal variation, demonstrating effective resource utilization and cost control. Despite these strengths, **the analysis identified customer service reliability as the primary operational challenge**. On-time delivery averaged only 44.6%, indicating that improvements in dispatch planning, route optimization, and delivery scheduling are needed to enhance service performance without increasing fleet capacity.

Data Note: Approximately 2% of Truck, Trailer, and Driver IDs were missing at random. These records were retained for aggregate operational analysis but excluded from entity-level performance comparisons where valid identifiers were required.

Key Operational Insights



Financial Performance

Revenue remained consistently higher than operating costs across the reporting period.

Key Finding:

Revenue declined to **\$20.16M** in February while operating cost reduced to **\$7.67M**, reflecting lower operational activity rather than improved cost efficiency.

Business Interpretation

The organization maintained healthy profitability, with operating costs largely moving in line with business activity.



Customer Demand

Key Finding:

Top 10 customers accounted for **4,812 deliveries (5.6% of 85,410)**, indicating a well-diversified customer portfolio.

Business Interpretation:

Customer demand is distributed across many accounts, reducing dependence on individual customers and lowering concentration risk.



Fleet Maintenance

Key Finding:



Fleet Operations

Fleet activity remained stable throughout the reporting period.

Key Finding:

Fleet activity (Trips) declined to **6,582** in February as downtime reduced to **4,765 hours**, suggesting reduced operational activity rather than maintenance improvement.

Business Interpretation:

Lower downtime should not automatically be interpreted as improved maintenance performance. Fleet availability was primarily influenced by changes in operational demand.



Facility Operations

Key Finding:

The busiest 10 facilities processed **17,609 loads (20.6% of total)**, with individual facility volumes ranging from **1,565 to 1,792 loads**.

Business Interpretation:

Operational workload is well balanced across the logistics network, indicating efficient distribution of processing capacity.



Safety Performance

Key Finding:

Preventive (**16.8%**) and Repair (**16.5%**) activities drive maintenance spending, while inspection activity remains lowest (**3.8%**).

Business Interpretation:

Maintenance investment is balanced across core activities. Increasing inspection frequency may support earlier fault detection and reduce future corrective maintenance.

The top 10 cities accounted for **71% of incidents**, with **68% classified as Moderate or Severe**, highlighting priority locations for safety improvements.

Business Interpretation:

Safety initiatives should focus on high-risk cities where higher-severity incidents are concentrated to maximize operational impact.

Truck Performance Investigation

A detailed truck-level analysis was conducted using fleet utilization, operating cost, downtime, revenue, trips completed, fuel efficiency, and incident records.

High-Performing Assets

Examples: [TRK00044](#), [TRK00039](#), [TRK00025](#).

These trucks consistently recorded:

- High utilization
- High trip volumes
- High revenue generation
- Higher operating costs
- Higher downtime

Notably, **TRK00044** generated the highest revenue while maintaining zero recorded incidents, demonstrating that high operating cost and downtime were associated with intensive operational use rather than poor performance.

Lower-Performing Assets

Examples: [TRK00060](#), [TRK00108](#), [TRK00018](#), [TRK00059](#), [TRK00013](#).

These trucks recorded:

- Lower operating costs
- Lower utilization
- Lower trip volumes
- Lower revenue generation

This suggests that lower operating costs were largely driven by reduced operational activity rather than superior cost efficiency.

Data Quality Assessment

The dataset was assessed before analysis to ensure the reliability of the reported insights.

Data Quality Findings:

Observation	Impact
1,672 (2%) missing Truck IDs	Randomly distributed across completed trips and excluded from truck-level analysis where necessary.
1,680 (2%) missing Trailer IDs	Randomly distributed and did not materially affect overall logistics performance analysis.
1,714 (2%) missing Driver IDs	Randomly distributed and excluded from driver-specific performance metrics where applicable.

All trip records were marked	Trip completion rate could not be meaningfully evaluated because there were no incomplete or cancelled trips in the dataset.
120 registered trucks vs. 92	Twenty-eight trucks recorded no operational activity during the reporting period and were treated as inactive assets in truck performance analysis.
Inactive trucks incurred	Indicates maintenance activities were recorded for non-operational assets, possibly representing reserve, parked, or workshop vehicles.

Data Quality Conclusion:

Despite a small proportion of randomly missing Truck, Trailer, and Driver IDs (approximately **2%** each), the missing values showed **no identifiable pattern by date, truck, driver, or operational status**, suggesting they are **Missing Completely at Random (MCAR)**. As a result, the overall dashboard insights and fleet-level analyses remain reliable.

However, driver- and truck-level investigations were conducted using only records with valid identifiers to ensure accurate performance comparisons.

Strategic Recommendations

1. Improve Delivery Reliability

Increase the On-Time Delivery Rate by reviewing dispatch schedules, route planning, loading processes, and delivery sequencing.

2. Optimize Maintenance Planning

Focus preventive maintenance and inspection efforts on high-utilization trucks to reduce downtime while maintaining operational output.

3. Strengthen Safety Management

Prioritize driver coaching, route assessments, and safety initiatives in cities with the highest concentration of Moderate and Severe incidents.

4. Sustain Operational Stability

Continue monitoring fleet utilization, operating costs, fuel efficiency, and customer demand to preserve the current level of operational performance.

Overall Conclusion

The logistics operation demonstrates **strong operational stability**, with consistent fleet utilization, balanced maintenance spending, stable fuel efficiency, and revenue that consistently exceeds operating costs. Customer demand and facility workloads are well distributed, reducing dependency on individual customers or locations.

The analysis also shows that the fleet's highest-performing trucks incur the greatest operating costs and downtime because they support the highest operational workload. Consequently, **maintenance optimization, rather than reduced asset utilization, offers the greatest opportunity for improving fleet availability.**

The most significant opportunity for business improvement remains **delivery reliability**. By strengthening dispatch planning, maintenance scheduling, and targeted safety initiatives, the organization can improve customer service while sustaining its current operational efficiency.

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